31/7 Recap

Probability space

→ Sample space S

→ Collection of events: subsets of S (usually power set)

- → S: event
- → A: event $\Rightarrow$ A^c: event
- → $A_1, A_2, \ldots$: events $\Rightarrow$ $A_1 \cup A_2 \cup \cdots$ event

→ Probability measure

P: Collection of events $\to \mathbb{R}$

Axioms: (1) $P(A) \geq 0$ for any event A.

(2) $P(S) = 1$

(3) $A_1, A_2, A_3, \ldots$: infinite sequence of disjoint events

$$P(A_1 \cup A_2 \cup A_3 \cup \cdots) = P(A_1) + P(A_2) + P(A_3) + \cdots$$

→ $P(\phi) = 0$

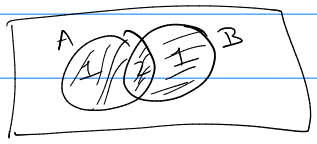
→ $P(A^c) = 1 - P(A)$

→ $B \subseteq A$ (events) $\quad P(B) \leq P(A)$



→ $0 \leq P(A) \leq 1$

→ $P(A) + P(B) = P(A \cup B) + P(A \cap B)$



→ Union bound $P(A \cup B) \leq P(A) + P(B)$

$A_1, A_2, \ldots$

$$P(A_1 \cup A_2 \cup A_3 \cup \ldots) \leq P(A_1) + P(A_2) + P(A_3) + \ldots$$

Finite sample spaces

Ex: ① Coin toss $S = \{H, T\}$

Σ_{events}: $\emptyset$, $\{H\}$, $\{T\}$, $\{H, T\}$

P: 0 $\quad p \in [0,1]$ $\quad 1-p$ $\quad 1$

② Throw a die $S = \{1, 2, 3, 4, 5, 6\}$

Σ_{events}: 64 ....

$\{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{6\}$

$p_1 \quad p_2 \quad p_3 \quad p_4 \quad p_5 \quad p_6$ } ***

$1 \geq p_i \geq 0 \qquad \sum_{i=1}^{6} p_i = 1$

Σ_{ent} $A = \{a_1, a_2, a_3\}$

$P(A) = p_{a_1} + p_{a_2} + p_{a_3}$

③ Draw a card from a shuffled deck

52 cards

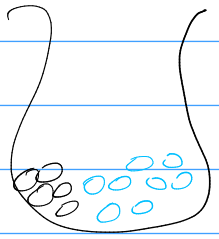
4 suits: clubs, diamonds, hearts, spades

each suit: 2, ..., 9, 10, J, Q, K, A

$S = \{2C, \ldots, AC, \ldots, 2S, \ldots, AS\}$

$1/52 \quad \ldots \quad 1/52 \quad \ldots \quad 1/52 \quad \ldots \quad 1/52$

4) Marbles in urns



5 black
8 blue

$S = \{k_1, k_2, \ldots, k_5, b_1, \ldots, b_8\}$ with probabilities $1/13, \ldots, 1/13$

Event $\text{Black} = \{k_1, \ldots, k_5\}$ $5/13$

$\text{Blue} = \{b_1, \ldots, b_8\}$ $8/13$

5) Birthday problem

n people in a room

$1, 2, \ldots, n$

$\{1, 2, \ldots, 365\}^n$

E: some 2 out of n share birthdays.

$$P(E) = 1 - \frac{365 \cdot 364 \cdots (365-n+1)}{365^n}$$

$$\left(1 - \frac{1}{365}\right)\left(1 - \frac{2}{365}\right) \cdots \left(1 - \frac{n-1}{365}\right)$$

$x \in [0,1]$

$1 - x \leq e^{-x}$

$\leq e^{\frac{-n^2}{2 \cdot 365}}$

Sample spaces with "uniform" outcomes

$S = \{s_1, s_2, \ldots, s_K\}$

$p \quad p \quad p \cdots p = 1/K$

$A = \{a_1, a_2, \ldots, a_m\}$ $\quad P(A) = \frac{m}{K}$

6) Throw a die twice

$S = \{(1,1), (1,2), \ldots, (6,6)\}$

$P(\text{sum is } 7) = \frac{6}{36}$

$P(\text{two are equal}) = \frac{30}{36} = \frac{36-6}{36}$

$$P(\text{two are unequal}) = 1 - P(\text{two equal})$$

(7) Throw a die thrice

$$P(\text{Some 2 are equal}) = 1 - \frac{6 \cdot 5 \cdot 4}{6^3}$$